

Report

Result

Size

Time

Cycles

GPU

SM Frequency

Process

Attributes

Current

optimized

572 - convolution3D_kernel

(24, 24, 1)x(32, 32, 1)

98.94 us

160,167

0 - NVIDIA GeForce RTX 4060 Laptop GPU

1.62 Ghz

[130715] conv3d_optimized

Baseline 1

baseline

572 - convolution3D_kernel

(24, 24, 1)x(32, 32, 1)

45.06 us

72,913

0 - NVIDIA GeForce RTX 4060 Laptop GPU

1.62 Ghz

[104238] conv3d_baseline

Summary

Details

Source

Context

Comments

Raw

Session

Compare

Tools

View

Export

GPU Speed of Light Throughput

GPU Throughput Chart

High-level overview of the throughput for compute and memory resources of the GPU. For each unit, the throughput reports the achieved percentage of utilization with respect to the theoretical maximum. Breakdowns show the throughput for each individual sub-metric of Compute and Memory to clearly identify the highest contributor. High-level overview of the utilization for compute and memory resources of the GPU presented as a roofline chart.

Compute (SM) Throughput [%]

28.87 (-1.90%)

Duration [us]

98.94 (+119.60%)

Memory Throughput [%]

28.00 (-54.52%)

Elapsed Cycles [cycle]

160,167 (+119.67%)

L1/TEX Cache Throughput [%]

27.34 (-15.80%)

SM Active Cycles [cycle]

150,034.12 (+127.04%)

L2 Cache Throughput [%]

8.70 (-54.21%)

SM Frequency [Ghz]

1.62 (-0.07%)

DRAM Throughput [%]

28.00 (-54.52%)

DRAM Frequency [Ghz]

7.99 (+0.16%)

Latency Issue

This workload exhibits low compute throughput and memory bandwidth utilization relative to the peak performance of this device. Achieved compute throughput and/or memory bandwidth below 60.0% of peak typically indicate latency issues. Look at [Scheduler Statistics](#) and [Warp State Statistics](#) for potential reasons.

Key Performance Indicators

Roofline Analysis

The ratio of peak float (FP32) to double (FP64) performance on this device is 64.1. The workload achieved 2% of this device's FP32 peak performance and 0% of its FP64 peak performance. See the [Profiling Guide](#) for more details on roofline analysis.

GPU Throughput

Compute (SM) [%]

Memory [%]

Speed of Light (SOL) [%]

Compute Workload Analysis

Pipe Utilization (Elapsed Cycles)

Detailed analysis of the compute resources of the streaming multiprocessors (SM), including the achieved instructions per clock (IPC) and the utilization of each available pipeline. Pipelines with very high utilization might limit the overall performance.

Executed Ipc Elapsed [inst/cycle]

1.15 (+95.58%)

SM Busy [%]

28.87 (+94.67%)

Executed Ipc Active [inst/cycle]

1.23 (+89.04%)

Issue Slots Busy [%]

28.87 (+94.67%)

Issued Ipc Active [inst/cycle]

1.23 (+88.16%)

Low Utilization

Est. Local Speedup: 81.16%

All compute pipelines are under-utilized. Either this workload is very small or it doesn't issue enough warps per scheduler. Check the [Launch Statistics](#) and [Scheduler Statistics](#) sections for further details.

Key Performance Indicators

Pipe Utilization (% of elapsed cycles)

Pipe Utilization (% of peak instructions executed over elapsed cycles)

ALU

FMA

FP64

Tensor (All)

Tensor (FP)

Tensor (INT)

LSU

ALU

FMA

CBU

ADU

Uniform

FMA (FP16)

FP64 (DMMA)

FP64 (FP64)

TEX

Tensor (FP)

Tensor (INT)

XU

Utilization [%]

Memory Workload Analysis

Detailed analysis of the memory resources of the GPU. Memory can become a limiting factor for the overall kernel performance when fully utilizing the involved hardware units (Mem Busy), exhausting the available communication bandwidth between those units (Max Bandwidth), or by reaching the maximum throughput of issuing memory instructions (Mem Pipes Busy). Detailed chart of the memory units.

Memory Throughput [Gbyte/s]

71.60 (-54.45%)

Mem Busy [%]

14.68 (-10.35%)

L1/TEX Hit Rate [%]

13.99 (-77.77%)

Max Bandwidth [%]

28.00 (-54.52%)

L2 Hit Rate [%]

46.67 (+11.45%)

Mem Pipes Busy [%]

25.64 (-12.89%)

L2 Compression Input Sectors [sector]

0 (+0.00%)

Local Memory Spilling Requests

0 (+0.00%)

L2 Compression Ratio

0 (+0.00%)

Local Memory Spilling Request Overhead [%]

0 (+0.00%)

L2 Compression Success Rate [%]

0 (+0.00%)

L2 Persisting Size [Mbyte]

6.29 (+0.00%)

Memory Chart

Values: Transfer Size Inactivity: Greyed Out

Kernel

Global

Local

Texture

Surface

Load Global Store Shared

Shared

Shared Memory

ICC

IMC

IDC

L2 Cache

L2 Compression Ratio

GCC

System Memory

Device Memory

Peer Memory

% Peak

Launch Statistics

Summary of the configuration used to launch the kernel. The launch configuration defines the size of the kernel grid, the division of the grid into blocks, and the GPU resources needed to execute the kernel. Choosing an efficient launch configuration maximizes device utilization.

Grid Size

576 (+0.00%)

Function Cache Configuration

CachePreferNone (CachePreferNone)

Registers Per Thread [register/thread]

25 (-21.88%)

Static Shared Memory Per Block [kbyte/block]

14.28 (+inf%)

Block Size

1,024 (+0.00%)

Dynamic Shared Memory Per Block [byte/block]

0 (+0.00%)

Threads [thread]

589,824 (+0.00%)

Driver Shared Memory Per Block [kbyte/block]

1.02 (+0.00%)

Waves Per SM

24 (+0.00%)

Shared Memory Configuration Size [Kbyte]

16.38 (+100.00%)

Uses Green Context

0 (+0.00%)

Stack Size

1,024 (+0.00%)

SMs [SM]

24 (+0.00%)

TPCs

12 (+0.00%)

Enabled TPC IDs

all (all)

-

Occupancy

% Occupancy Graphs

Occupancy is the ratio of the number of active warps per multiprocessor to the maximum number of possible active warps. Another way to view occupancy is the percentage of the hardware's ability to process warps that is actively in use. Higher occupancy does not always result in higher performance, however, low occupancy always reduces the ability to hide latencies, resulting in overall performance degradation. Large discrepancies between the theoretical and the achieved occupancy during execution typically indicates highly imbalanced workloads.

Theoretical Occupancy [%]

66.67 (+0.00%)

Block Limit Registers [block]

2 (+0.00%)

Theoretical Active Warps per SM [warp]

32 (+0.00%)

Block Limit Shared Mem [block]

1 (-87.50%)

Achieved Occupancy [%]

65.87 (+15.22%)

Block Limit Warps [block]

1 (+0.00%)

Achieved Active Warps Per SM [warp]

31.62 (+15.22%)

Block Limit SM [block]

24 (+0.00%)

Theoretical Occupancy

Est. Local Speedup: 33.33%

This kernel has uncoalesced global accesses according to its occupancy are below the hardware maximum of 12. This kernel's theoretical occupancy (66.7%) is limited by the required amount of shared memory, and the number of warps within each block.

Key Performance Indicators

Impact of Varying Register Count Per Thread

Impact of Varying Block Size

Impact of Varying Shared Memory Usage Per Block

% Occupancy

Registers Per Thread

Block Size

Shared Memory Per Block

GPU and Memory Workload Distribution

Analysis of workload distribution in active cycles of SM, SMP, SMSP, L1 & L2 caches, and DRAM

Average SM Active Cycles [cycle]

150,034.12 (+127.04%)

Average L1 Active Cycles [cycle]

150,034.12 (+127.04%)

Average L2 Active Cycles [cycle]

147,685.38 (+140.44%)

Average SMSP Active Cycles [cycle]

149,374.82 (+134.93%)

Average DRAM Active Cycles [cycle]

221,372 (+0.03%)

Total SM Elapsed Cycles [cycle]

3,839,608 (+119.44%)

Total L2 Elapsed Cycles [cycle]

2,545,232 (+118.48%)

Total SMSP Elapsed Cycles [cycle]

15,358,432 (+119.44%)

Total DRAM Elapsed Cycles [cycle]

3,162,112 (+119.94%)

Workload Distribution

SM Active Cycles

SMSP Active Cycles

L1 Active Cycles

L2 Active Cycles

DRAM Active Cycles

Average

Min

Max

Sum

Source Counters

Source metrics, including branch efficiency and sampled warp stall reasons. Warp Stall Sampling metrics are periodically sampled over the kernel runtime. They indicate when warps were stalled and couldn't be scheduled. See the documentation for a description of all stall reasons. Only focus on stalls if the schedulers fail to issue every cycle.

Branch Instructions [inst]

776,676 (+2,009.62%)

Branch Efficiency [%]

85.69 (+inf%)

Branch Instructions Ratio [%]

0.18 (+391.54%)

Avg. Divergent Branches [branches]

662.88 (+inf%)

Uncoalesced Global Accesses

Est. Speedup: 21.78%

This kernel has uncoalesced global accesses resulting in a total of 112332 excessive sectors (23% of the total 478908 sectors). Check the L2 Theoretical Sectors Global Excessive table for the primary source locations. The [CUDA Programming Guide](#) has additional information on reducing uncoalesced device memory accesses.

Key Performance Indicators

L2 Theoretical Sectors Global Excessive

Location

Value

Value (%)

0x759197272d70 in convolution3D_kernel

37,444

33

0x7591972729e0 in convolution3D_kernel

37,444

33

0x759197272680 in convolution3D_kernel

37,444

33

0x7591972730e0 in convolution3D_kernel

0

0

Warp Stall Sampling (All Samples)

Most Instructions Executed

Location

Value

Value (%)

Location

Value

Value (%)

0x759197272ee0 in convolution3D_ke_

926

27

0x759197272d80 in convolution3D_ke_

40,796

1

0x7591972726a0 in convolution3D_ke_

437

13

0x7591972729f0 in convolution3D_ke_

40,796

1

0x759197272a00 in convolution3D_ke_

423

12

0x759197272690 in convolution3D_ke_

40,796

1

0x759197272d90 in convolution3D_ke_

414

12

0x759197272dd0 in convolution3D_ke_

39,168

1

0x759197273100 in convolution3D_ke_

59

2

0x759197272dc0 in convolution3D_ke_

39,168

1

Still missing data? Collect the [full/metric set](#), enable [individual sections](#) or learn how to select [individual metrics](#) for collection.